

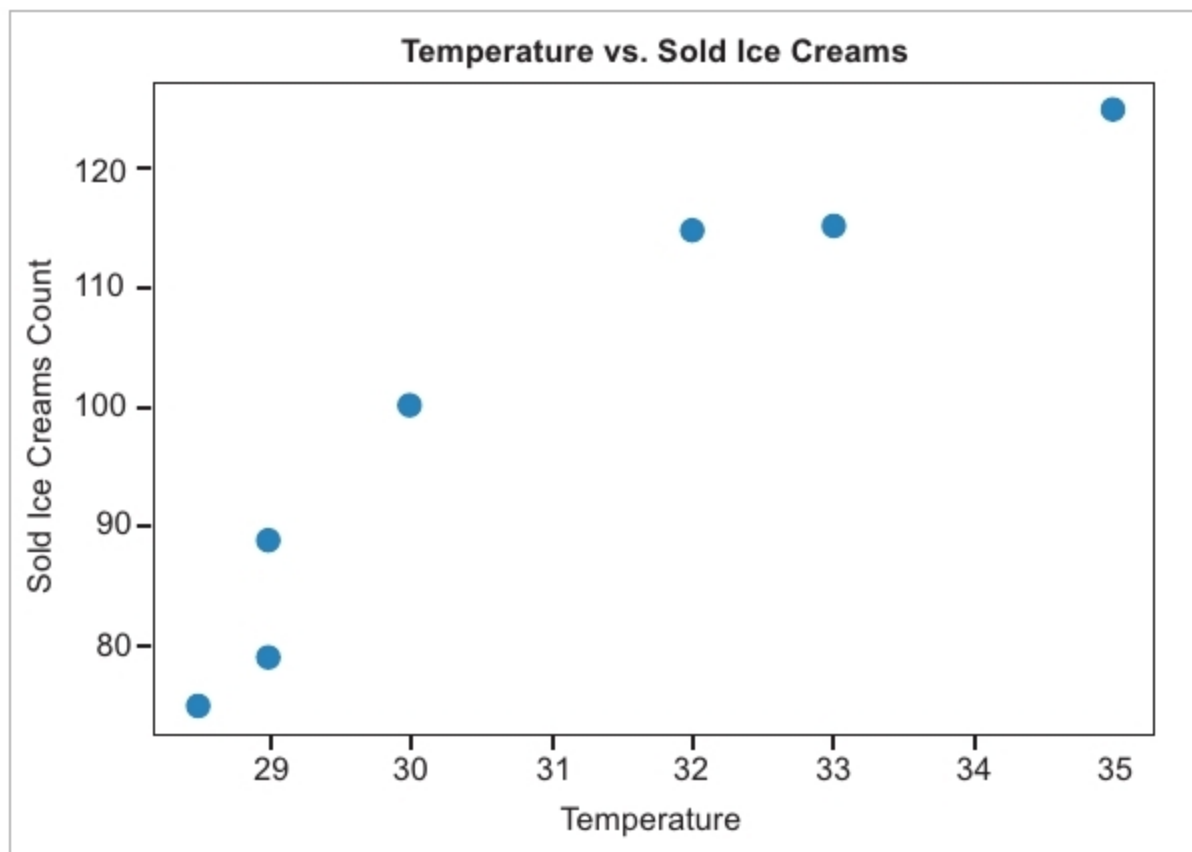


Figure 3: Plotting data with a scatter plot using Matplotlib

scatter plot with Matplotlib, we can use the `plt.scatter()` function. In this, the first argument is used for the data on the horizontal axis and the second for the vertical axis. The results are plotted as a scatter plot using the following Python code:

```
import matplotlib.pyplot as plt
temp = [30, 32, 33, 28.5, 35, 29, 29]
ice_creams_count = [100, 115, 115, 75, 125, 79, 89]
plt.scatter(temp, ice_creams_count)
plt.title("Temperature vs. Sold Ice Creams")
plt.xlabel("Temperature")
plt.ylabel("Sold Ice Creams Count")
plt.show()
```

4) Creating a histogram:

A histogram divides the values within a numerical variable into 'bins' and counts the number of observations that fall into each bin. In Matplotlib, a histogram can be created using the `hist` method:

```
import matplotlib.pyplot as plot
import numpy as np
# Random number generation from 0 to 100
x1 = np.random.random_
integers(0,100,100)
# Plot frequency distribution using histogram
plot.hist(x1, facecolor="r",
edgecolor='b')
plot.title("Frequency distribution
Histogram - Values from 0-100")
plot.margins(0)
plot.xticks(range(0,110,10))
# Display the histogram
plot.show()
```

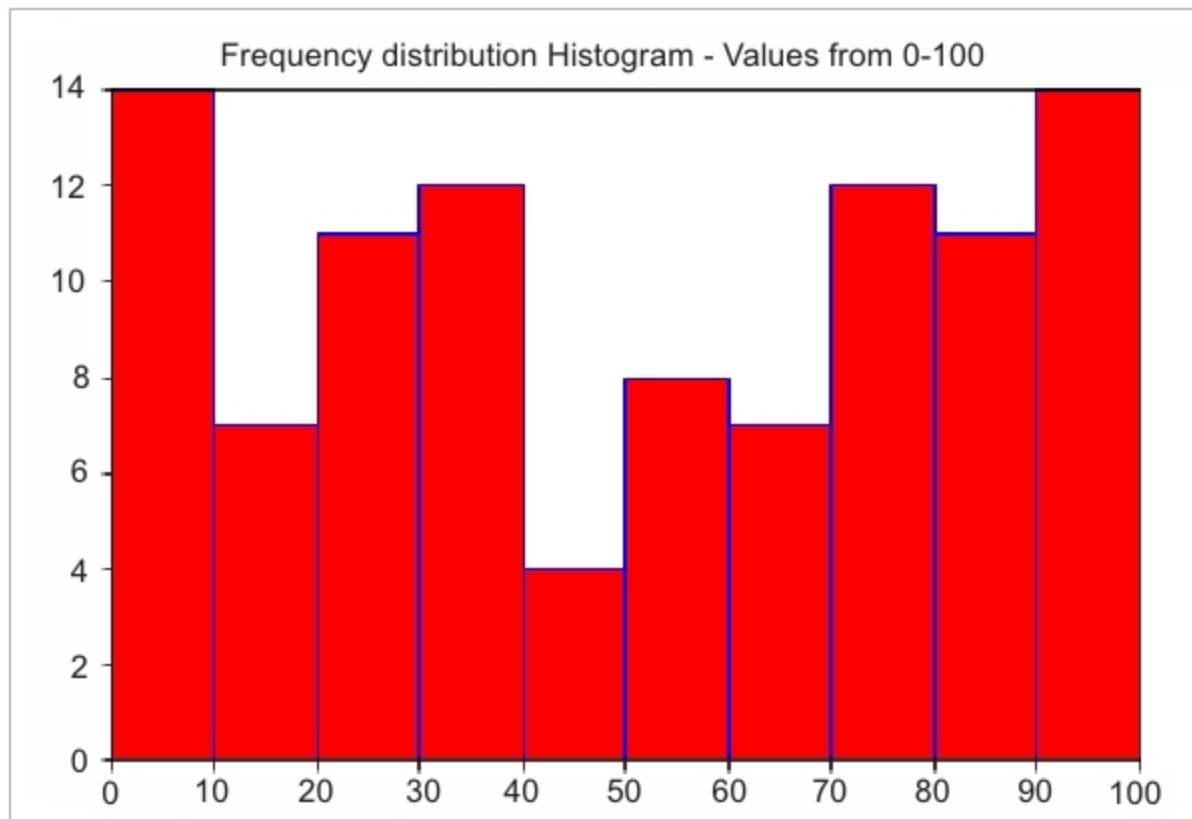


Figure 4: Plotting a histogram using Matplotlib

method. The bar chart is useful for categorical data that doesn't have a lot of different categories.

```
# importing the matplotlib library
import numpy as np
import matplotlib.pyplot as plt
# The alias 'plt' has been set for
simplification purpose
# creating the dataset
data={'Cloud':20, 'IoT':15, 'Big
```

```
Data':30, 'ML':10}
names=list(data.keys())
values=list(data.values())
fig=plt.figure(figsize=(7,5))
#Creating the bar plot
plt.bar(names, values)
plt.show()
```

3) Plotting data with a scatter plot:

A scatter plot shows the data as a collection of points. To make a

5) Visualisation using a pie chart:

Pie charts show proportions and percentages between categories, by dividing a circle into proportional segments. In Matplotlib, a pie chart can be created using the `pie()` function: